

MA341

APPLIED ODE'S

Homework 4.3

1. Find general solution to the following equations:

(a) $x^2y'' - 3xy' + 4y = 0$

(b) $x^2y'' + xy' + 4y = 0$

(c) $x^2y'' + 4xy' + 5y = 0$

(d) $x^2y''' = 2y'$. Hint: Use substitution $w = y'$.

2. Use method of undetermined coefficients to find general solution to the following equations:

(a) $x^2y'' - 2y = \sin \ln |x|$

(b) $(x - 2)^2y'' - 3(x - 2)y' + 4y = x$. Hint: use change of variables $t = x - 2$.

(c) $y'' - 4y = x^2 + e^x$

(d) $y'' + y = \sin x$

(e) $y'' + 2y' - 3y = 2 + 2\cos^2 x$. Hint: use trigonometric identities.

3. Use method of variation of parameter to find general solution to the following equations:

(a) $y'' + y = \tan x$

(b) $y'' + 4y = \cot^2 2x$

(c) $y'' + 12y' + 32y = \frac{16}{1+e^{4x}}$